predict_spatial_resolution_grid_age.R

June 18, 2026

Contents

`predict_spatial_resolution_grid_age`
Predict Spatial-Resolution Grid for One Age

Description

Builds abiotic and optional spatial predictors for one age slice and returns long-format taxon occurrence probabilities.

Usage

```
predict_spatial_resolution_grid_age(
  prediction_inputs,
  data_grid,
  data_grid_coords_projected,
  age,
  abiotic_variables,
  x_lim,
  y_lim,
  cache_dir,
  spatial_mode = "spatiotemporal",
  climate_fn = NULL,
  spatial_interpolate_fn = NULL,
  spatiotemporal_interpolate_fn = NULL,
  predict_fn = NULL
)
```

Arguments

`prediction_inputs`
Named list returned by [read_spatial_resolution_prediction_inputs()].

`data_grid` Prediction grid with ‘grid_id‘, ‘coord_long‘, and ‘coord_lat‘.

data_grid_coords_projected
: Projected prediction grid coordinates with row names ‘grid_<id>‘.

age
: Numeric scalar age in years before present.

abiotic_variables
: Character vector of CHELSA variables to extract.

x_lim
: Numeric longitude range.

y_lim
: Numeric latitude range.

cache_dir
: Existing CHELSA cache directory.

spatial_mode
: Spatial predictor mode: ‘"spatial"‘, ‘"spatiotemporal"‘, or ‘"none"‘.

climate_fn
: Climate extraction function. If ‘NULL‘ (default), [extract_prediction_climate()] is used.

spatial_interpolate_fn
: Spatial interpolation function. If ‘NULL‘ (default), [interpolate_mev_to_grid()] is used.

spatiotemporal_interpolate_fn
: Spatiotemporal interpolation function. If ‘NULL‘ (default), [interpolate_st_mev_to_grid()] is used.

predict_fn
: Prediction function. If ‘NULL‘ (default), the function resolves ‘sjSDM:::predict.sjSDM‘.

Value

Long tibble with predicted probabilities by grid cell, age, and taxon.

Examples

```
## Not run:
predict_spatial_resolution_grid_age(
  prediction_inputs = prediction_inputs,
  data_grid = data_grid,
  data_grid_coords_projected = data_grid_coords_projected,
  age = 0,
  abiotic_variables = c("bio1", "bio12"),
  x_lim = c(-10, 40),
  y_lim = c(35, 70),
  cache_dir = "Data/Temp/chelsa/prediction"
)

## End(Not run)
```

Index